MUZI CONSOLE

A console strip with an engineer that listens first.

User manual · version 0.1



Channel & Bus

One plugin for a single track or for the whole mix.

E & G curves

Two console EQ characters, inspired by classic British desks.

Listen & Suggest

Hears the track, shows its reasons, changes nothing until you press APPLY.

01 Start here

Muzi Console is a channel strip you can put on every track, and on your buses and master. It also has an engineer built in. The engineer listens to the track, compares it with a target for that kind of sound, and suggests settings with a reason for each move. You stay in charge: nothing changes until you press APPLY.

Open the project

1. Unzip **Muzi-Console-v0.1-Projucer.zip** (or download the repo from GitHub) and double-click **Muzi-Console.jucer**. JUCE 9.0.5 is already inside, in the **JUCE** folder, so nothing else needs installing.
2. In Projucer, click the Xcode button. You can also open **Builds/MacOSX/Muzi Console.xcodeproj** directly.
3. Press Build. Debug and Release are both optimised, so the Debug build Xcode makes when you press Run plays smoothly on an Intel Mac. You get a VST3, an AU and a Standalone app, and the plugins are copied into **~/Library/Audio/Plug-Ins**.
4. In FL Studio: **Options › Manage plugins › Find installed plugins**, then load **Muzi Console** on a mixer insert.

Your first two minutes

1. Put Muzi Console on a vocal and leave it in **CHANNEL** mode.
2. In the engineer panel, pick **Vocal** and your genre, for example **Trap**.
3. Press **LISTEN**, then play the busiest part of the song. It listens until it has heard 10 seconds of sound. Silence doesn't count.
4. Read the moves, and look at the display: gold solid is the suggested EQ, teal is the predicted result.
5. Press **APPLY**. Compare with bypass at the same loudness, then adjust anything by ear. **UNDO** puts every control back.



While it listens, the display draws what it's hearing in real time and the bar fills toward 10 seconds. Press STOP to finish early (it needs at least 2 seconds of sound).

THE ONE RULE

The engineer gives you a starting point, not a verdict. Its targets are general shapes for each kind of sound. Your song, your arrangement and your taste come first. If a move sounds wrong, it is wrong.

02 Panel tour



1. **CHANNEL / BUS**: which job the strip is doing. It changes the dynamics section, the EQ guard and how the engineer thinks.
2. **CURVE E / G**: the EQ's character (chapter 4). It changes the shape of every bell and shelf, not the settings.
3. **The glass display**: the live EQ curve (white), plus what the engineer heard, its target and the predicted result.
4. **Band nodes**: drag to set frequency and gain. Scroll on a mid node to change its width. Double-click a node to reset its gain.
5. **Role and genre**: what's on the channel and the style. Change them after a listen and the suggestion updates without listening again.
6. **LISTEN**: starts a 10-second listen. While it runs, the button shows the time and works as STOP.
7. **The engineer's notes**: every move with its reason, plus warnings such as "fix this on the channel".
8. **APPLY / UNDO**: APPLY sets the suggestion. UNDO puts back every control as it was before APPLY.
9. **Filters**: high-pass (18 dB/oct) and low-pass (12 dB/oct), each with its own IN button.
10. **EQ**: four colour-coded bands. LF and HF switch between shelf and BELL. LMF and HMF have WIDTH (Q).
11. **Dynamics**: channel compressor and gate in CHANNEL mode. It becomes the bus glue compressor in BUS mode.
12. **Output**: TRIM (input gain), DRIVE (console colour), OUTPUT, and the IN / GR / OUT meters. The tick on IN and OUT marks -18 dBFS.

03 Two jobs, one plugin

On a real console, a channel strip and the mix bus do different jobs. A channel can be bold: big cuts, fast compression, a gate. A bus should be gentle: small EQ moves and a few dB of glue. Muzi Console switches between the two.

CHANNEL



BUS



The engineer always listens to the raw input, before TRIM, so its trim suggestion is measured from where your track actually is.

CHANNEL: USE IT FOR

Vocals, kicks, 808s, snares, samples, any single source. The engineer sets gain staging, filters, all four bands, the compressor and, for 808s in Trap and Drill, some drive.

BUS: USE IT FOR

Drum and music buses, and the master. EQ moves are held to ±2 dB while **GUARD ±2** is lit. The engineer never trims a bus and only high-passes it at 20 Hz when there's sub-sonic junk.

The ±2 dB bus rule

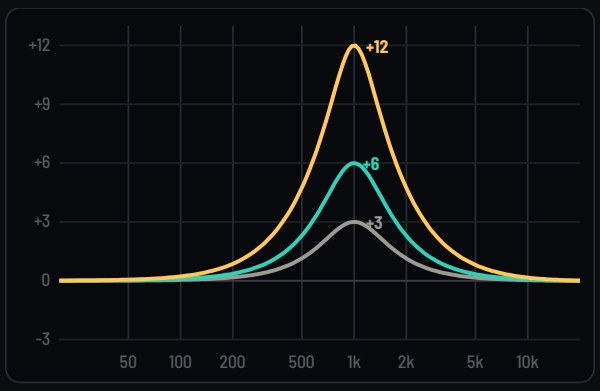
If a mix needs a bigger EQ move than 2 dB, the problem usually belongs to one track, not to the whole mix. Boosting 6 dB of sub on the master also lifts every vocal breath and hi-hat bleed in that range. So in BUS mode the engineer holds its moves to 2 dB, tells you how much the mix really wanted, and names the likely culprit:

> Around 45 Hz the mix wants +6.0 dB - too much for a bus. Fix it at the source: likely the 808 or kick. (Bus move held to +2.0 dB.)

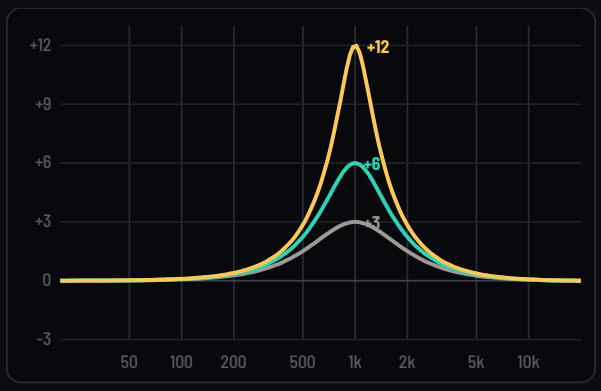
You can switch GUARD off to make bigger bus moves yourself. The engineer still keeps its own suggestions within 2 dB.

04 The E and G curves

Classic British consoles came in two EQ temperaments. One kept its bells the same width however hard you pushed them. The other narrowed them as you boosted. Muzi Console gives you both. They're inspired by those desks, not copies of any unit. Switching between E and G keeps every knob where it is and changes only the shape of the curves.



E: +3, +6 and +12 dB at 1 kHz, width 1.00. The bell keeps the same shape at every gain. At +12 dB it still lifts 2 kHz by about 5 dB.

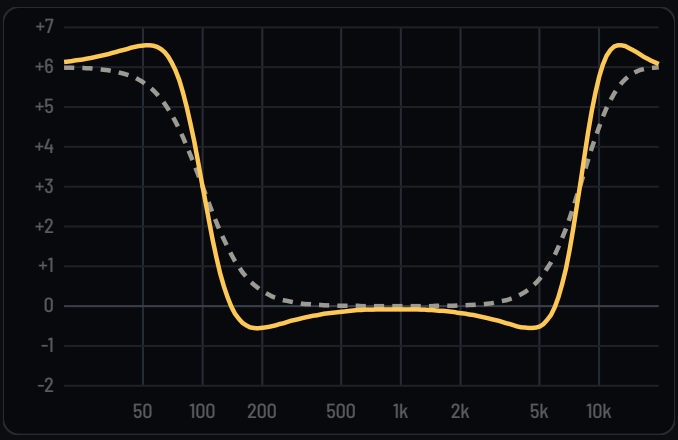


G: the same three moves. Small boosts are broad and gentle, and big boosts narrow in on the frequency. At +12 dB, 2 kHz only rises about 3 dB.

The G law, in numbers

On the G curve, the effective width of a bell is the WIDTH knob times a factor that depends on the gain:

GAIN (EITHER DIRECTION)	3 DB	6 DB	12 DB	15 DB
G width factor	0.66 (broad)	0.87	1.29	1.50 (focused)
E width factor	0.85	0.85	0.85	0.85



Shelves

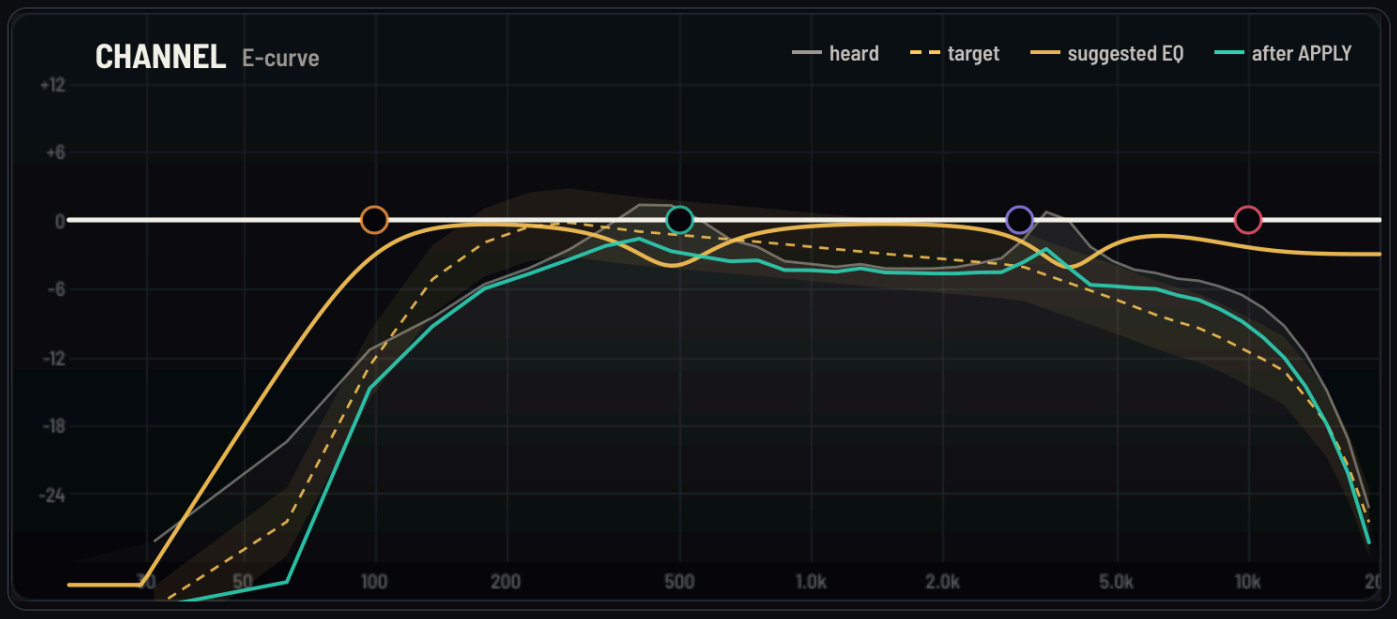
The plot shows +6 dB shelves at 100 Hz and 8 kHz. **E** shelves (dashed) are smooth and gentle. **G** shelves (solid) are steeper and carry a small dip of about half a dB just inside the turnover. That dip is what makes a G low boost sound tight rather than woolly.

Which one?

- **E**: broad tone shaping, vocals, buses, anything where you want the EQ to stay out of the way.
- **G**: drums, 808s, and surgical boosts that need to be focused. Also gentle broad lifts with small gains.

All curves on this page are computed from the plugin's own EQ code at 48 kHz, not drawn by hand.

05 Reading the display



YOU SEE	IT MEANS
White curve	The EQ that's running now: filters plus all four bands, in the current curve flavour.
Grey area (heard)	What the engineer heard, as the level in each of its 40 hearing bands. The ear works in critical bands, so the display does too.
Dashed gold + tube (target)	Where this role and genre usually sit, with a ±3 dB tolerance tube. Anything inside the tube is left alone.
Solid gold (suggested EQ)	The EQ the engineer proposes. It's only shown before you press APPLY. After that it's the same as the white curve.
Teal (after / after APPLY)	The predicted result: what you'd hear through the suggested EQ (or the live EQ, once applied).
L LM HM H nodes	The four bands. A filled node is active, a hollow node is at 0 dB. Blue nodes on the 0 dB line are the HPF and LPF, shown when they're IN.

Using the nodes

- **Drag** a node to set its frequency (left/right) and gain (up/down). Filter nodes move left and right only.
- **Scroll** over the LM or HM node to make the bell narrower or wider.
- **Double-click** a node to set its gain back to 0 dB. Double-click any knob to return it to its default.
- Hover over a node to read its exact frequency, gain and width.

The display is drawn once into a cached background and only redraws when something actually changes, which keeps it light on an Intel Mac. The meters update on their own small area.

06 Controls reference

CONTROL	RANGE (DEFAULT)	WHAT IT DOES
TRIM	±24 dB (0)	Input gain before everything else. Aim for an average of about -18 dBFS: the IN meter's tick.
HPF	16-350 Hz (80 Hz, out)	18 dB/octave high-pass. Steep enough to clear rumble, smooth enough not to ring.
LPF	3-22 kHz (18 kHz, out)	12 dB/octave low-pass, for taming fizz or pushing a sound back.
LF	±15 dB, 30-450 Hz (100 Hz)	Low shelf, or a bell with BELL lit. Weight, warmth, or cleaning out boom.
LMF	±15 dB, 200 Hz-2.5 kHz (500 Hz), width 0.5-2.5 (1.0)	Low-mid bell. Mud (200-400 Hz) and boxiness (400 Hz-1 kHz) live here.
HMF	±15 dB, 600 Hz-7 kHz (3 kHz), width 0.5-2.5 (1.0)	High-mid bell. Presence, bite and harshness.
HF	±15 dB, 1.5-16 kHz (10 kHz)	High shelf, or a bell with BELL lit. Air, brightness, or calming a fizzy top.
EQ IN	(in)	Takes the four bands out. The filters stay in.
GUARD ±2	BUS only (on)	Holds every band to ±2 dB in BUS mode.
DRIVE	0-100% (0)	Console colour: gentle 2nd and 3rd harmonics, level-matched so it doesn't just sound louder. Fully in by 50%, pushed harder above that.
OUTPUT	±24 dB (0)	Final level, after everything.

WHY -18 DBFS?

Analog consoles were designed around a nominal level, and so is this one. At about -18 dBFS average, the compressor thresholds and the DRIVE amount do what their numbers say: 30% drive is a touch of colour, not distortion. Set TRIM first (the engineer does it for you), mix with the faders, and leave OUTPUT for level-matching against bypass.

SHELF OR BELL?

LF and HF are shelves by default: they lift or lower everything below (LF) or above (HF) their frequency. Light **BELL** to make them peaks instead, for example a 60 Hz bell to add kick weight without lifting the 30 Hz rumble under it.

07 Dynamics reference

CHANNEL mode shows the compressor and gate. BUS mode replaces them with the glue compressor. Both feed the GR meter in the Output section.

Channel dynamics

THRESH	-40–0 dB (0)	Where compression starts. Soft 6 dB knee, peak-sensing, stereo-linked. Make-up gain is automatic, at half of what the compressor would take off a 0 dBFS peak.
RATIO	1:1–20:1 (4:1)	How hard it pushes back. At 1:1 the compressor is off.
RELEASE	0.1–4 s (0.3 s)	How fast it lets go.
FAST	(off)	Attack drops from 3 ms (lets the front of each hit through) to 0.3 ms (catches it).
GATE	-80 (Off)–0 dB	Gate/expander threshold. It opens in 0.5 ms, closes over 150 ms, and has 4 dB of hysteresis so it doesn't chatter.
RANGE	0–40 dB (40)	How far the gate turns things down when closed. 6–12 dB gives a gentle expander.
PRE-EQ / DYN IN	(off / in)	PRE-EQ puts the dynamics before the EQ. DYN IN bypasses compressor and gate together.

Bus compressor

THRESH / RATIO	-30–0 dB (0) / 2, 4, 10:1 (4:1)	Soft 4 dB knee. 2:1 for gentle glue, 4:1 for punch, 10:1 to squash.
ATTACK	0.1, 0.3, 1, 3, 10, 30 ms (3 ms)	Slow attacks (10–30 ms) let drum transients through, so the mix punches.
RELEASE	0.1, 0.3, 0.6, 1.2 s, Auto (Auto)	Auto: short peaks recover fast, sustained squeeze recovers slowly, so it follows the groove.
MAKE-UP / MIX	0–15 dB (0) / 0–100% (100%)	MIX below 100% blends in the uncompressed mix (parallel compression).
SC HPF	(off)	The compressor stops listening below 60 Hz, so a big 808 doesn't make the whole mix pump.

08 How the engineer decides

The engineer is rule-based, so every decision can be explained. It doesn't guess, and it doesn't copy a famous engineer's settings. It measures, compares, and applies the habits most mixers share.

1. Listen

It hears the raw track (before TRIM) on a worker thread, never in the audio callback. Only sound counts toward the 10 seconds: silence and quiet tails are skipped, and the analysis uses the loud, musical part (within 20 dB of the loudest moments).

2. Measure like an ear

The spectrum is grouped into **40 critical bands** from 20 Hz to 20 kHz, one per ERB, the ear's own frequency resolution. Levels are per octave, so pink noise reads flat.

3. Compare with the target

Each role has a target shape, and the genre shifts it. The engineer lines your track up with the target on the bands that matter for that role, so one big resonance can't throw the comparison off.

4. Cuts first

For mud and boxiness (LMF) and harshness (HMF), it looks for two things: **excess** over the target, and **resonances** that stick up above their own octave neighbourhood even where the target slopes. It fits a bell to each problem and uses the fitted width, so a narrow ring gets a narrow cut and a broad hump gets a broad one.

5. Then boosts, sparingly

It only boosts (low weight, body, presence, air) where the role wants it and the track is clearly short of the target.

6. Restraint

Anything within 3 dB of the target (2 dB on a bus) is left alone. It cuts 75% of an excess (up to 9 dB) and boosts 60% of a shortfall (up to 6 dB, presence up to 5 dB), in half-dB steps.

Gain and dynamics

- **Trim** (channel only): brings the average level to about -18 dBFS, where the compressor and drive are calibrated.
- **Compressor**: the engineer measures the track's peaks every 10 ms and solves for the threshold that gives the role's goal. That's about 5 dB of gain reduction on a vocal and 3 dB on a kick, measured on your track rather than guessed.
- **Bus**: 2:1 for gentle genres, 4:1 for Trap, Drill and Boom Bap. 30 ms attack, Auto release, about 2 dB of glue, and SC HPF on when the low end is heavy or the genre is Trap or Drill.

WHAT IT WON'T DO

It doesn't judge arrangement, panning, reverb or level between tracks, and it can't know what you're going for. A dark, lo-fi vocal might be exactly right for your song. Use the role and genre to tell it what you want, and ignore any suggestion that fights the vibe.

09 Roles and genres

ROLE	HIGH-PASS	WHERE IT ADDS	COMPRESSOR
Vocal	80 Hz (100 Hz in Trap / Drill)	Body 220 Hz if thin, presence 2.8 kHz (1.5 kHz in Pop), air	4:1, ~5 dB, 0.25 s
Kick	25 Hz	Low bell at 60 Hz (55 Hz in Trap / Drill), click 3.5 kHz	4:1, ~3 dB, 0.10 s
808 / Sub	None, the sub is the point	Low shelf 60 Hz. Drive 35% in Trap / Drill	3:1, ~3 dB, 0.20 s
Bass	30 Hz	Low shelf 90 Hz, definition 800 Hz	4:1, ~4 dB, 0.20 s
Snare / Clap	80 Hz	Body 220 Hz if thin, crack 4 kHz, air	4:1, ~4 dB, 0.15 s
Hats / Cymbals	400 Hz	Air only	2:1, ~2 dB, fast
Guitar	90 Hz	Body, presence 3 kHz, air	3:1, ~3 dB, 0.20 s
Keys / Synth	100 Hz	Presence 2.5 kHz, air	2:1, ~2 dB, 0.30 s
Sample / Loop	40 Hz	Low shelf 70 Hz, air	2:1, ~2 dB, 0.30 s
Mix Bus	Only 20 Hz, if sub-sonic junk	Low shelf 45 Hz, air (all held to ±2 dB)	Bus comp, ~2 dB glue

Picking Mix Bus while in CHANNEL mode switches the strip to BUS when you press APPLY.

GENRE	HOW IT SHIFTS THE TARGET
Neutral	The role's target as it is.
Trap	About +2 dB more sub below 60 Hz on low sources (kick, 808, bass, samples, mix), +1 dB of extra air. Vocal high-pass at 100 Hz. Bus at 4:1 with SC HPF.
Drill	Same extra sub as Trap, but a darker top (-1.5 dB above about 6 kHz).
R&B	Glossier top (+1.5 dB above about 9 kHz, air shelf at 12 kHz) and a slightly cleaner 250-500 Hz.
Boom Bap	Warmer 200-600 Hz (+1 dB) and dustier highs (-2 dB above about 10 kHz). Bus at 4:1.
Pop	+1 dB of presence around 1-3 kHz. Vocal presence moves to 1.5 kHz, for a forward, slightly nasal poke.

These targets are starting points drawn from common practice for each kind of sound. They are not measurements of any particular record.

10 Recipes and tips

RAP VOCAL THAT CUTS THROUGH A TRAP BEAT

CHANNEL, E curve, Vocal + Trap, LISTEN on the hook, APPLY. Then switch to **G** and nudge HMF up 1-2 dB around 3 kHz: G keeps a small boost broad and smooth. If the hi-hats then clash with the vocal's air, use Epsilon Carve on the beat.

808 THAT READS ON A PHONE

CHANNEL, 808 / Sub + Trap. The engineer leaves the sub alone (no high-pass) and adds 35% DRIVE, which creates harmonics an octave up so small speakers can hear the note. Check on laptop speakers: if you can hear the melody, it's working.

GLUE ON THE MASTER

BUS, Mix Bus + your genre, LISTEN on the drop. Read any "fix it at the source" notes first, because those are worth more than the bus EQ. Aim for 1-3 dB on the GR meter. Try MIX at 60-80% for a more open sound.

TIGHT DRUMS BUS

BUS, G curve, ATTACK 10-30 ms, RELEASE Auto, RATIO 4:1, about 3 dB GR. SC HPF on if the kick's sub makes it pump.

TAMING A RING YOU CAN HEAR

Drag the LM or HM node up +10 dB on the G curve (it narrows as it rises) and sweep until the ring jumps out. Then pull it down to a cut and scroll to narrow it. Or just LISTEN, because finding rings is what the resonance check does.

GENTLE EXPANDER ON A NOISY VOCAL

GATE just above the noise floor between phrases, RANGE 8-10 dB instead of 40. The breaths drop back without the vocal sounding chopped.

Good habits

- Compare with bypass at the **same loudness**. Use OUTPUT to match levels, because louder always sounds "better" at first.
- LISTEN to the busiest part of the song, not the intro. The engineer measures what it hears.
- Change role or genre after a listen to see how the suggestion changes. There's no need to listen again.
- If a bus note says "fix it at the source", put a second Muzi Console on that channel and LISTEN there.

11 Under the hood

ITEM	DETAIL
Formats	VST3, AU, Standalone (macOS, universal Intel + Apple Silicon from the GitHub build). VST3 and Standalone on Windows and Linux from the same project.
Latency	Zero. Safe while recording and monitoring live.
Channels	Mono or stereo. Stereo is processed linked: one gain decision for both sides.
EQ	Double-precision biquads. Changes glide over about 20 ms (frequencies glide on a log scale) so automation and dragging never zip.
Engineer analysis	8192-point FFT, Hann window, 50% overlap, grouped into 40 ERB bands. Runs on its own low-priority thread.
Sessions	All settings, plus role and genre, are saved with your session. Parameter IDs are fixed from v0.1, so sessions will keep loading in later versions.
Window	Resizable from 75% to 200%, keeping its proportions.

Tested before release

- **43 engine and engineer checks**. They cover the E/G curve laws, filter slopes and the bus guard; compressor, gate and drive levels; nulling when flat, and staying finite under abuse. They also plant problems in test audio and confirm the engineer finds them, that its cut actually flattens them, that it invents nothing on a clean source, and that it gives the right bus diagnostics.
- **Speed**: a minute of stereo with everything on and a sweeping EQ renders in about 0.36 s, roughly 165× real time.
- **pluginval strictness 8**, at 44.1, 48 and 96 kHz and block sizes 64-1024, including the editor: passed on the Release and Debug builds, with no JUCE assertions.

Troubleshooting

Xcode: "JUCE modules not found"	You opened a copy without the **JUCE** folder (for example GitHub's green Code › Download ZIP before JUCE was added). Use the Projucer zip or a full clone.
"Damaged" or blocked plugin	Builds downloaded from GitHub aren't notarised. In Terminal: **xattr -dr com.apple.quarantine** followed by the plugin's path (see INSTALL.txt).
FL Studio doesn't list it	Options › Manage plugins › Find installed plugins, and check the VST3 is in **~/Library/Audio/Plug-Ins/VST3**.
"Didn't hear enough"	Press play while it's listening: it needs at least 2 seconds of actual sound.